

ShopFlow AWS Project

EC2 Design

Virtual Machine - Spring Boot Host - SSM Session Manager

Module 8 of 18

Field	Value
Version	1.0
Date	June 2026
Region	ap-south-1 (Mumbai)
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Application	ShopVerse -- Spring Boot 3.2.5 / Java 21

1. Overview

Amazon EC2 provides the virtual machine that runs the ShopVerse Spring Boot application. The instance is placed in a private subnet with no public IP. Access is via AWS SSM Session Manager -- no SSH key pairs or port 22 required. The instance pulls its JAR from S3 and config from SSM/Secrets Manager at first boot.

2. Instance Configuration

Setting	Value	Reason
Instance Type	t3.small	2 GB RAM -- minimum for Spring Boot JVM
AMI	Amazon Linux 2023	AWS-managed, Java 21 Corretto available
AMI Lookup	SSM /aws/service/ami-amazon-linux-latest/...	Auto-resolves latest AMI -- no hardcoded ID
JDK	Amazon Corretto 21	AWS-supported OpenJDK, free
JVM Heap	-Xms256m -Xmx512m	Fits within 2 GB instance RAM
Root Volume	30 GB gp3 SSD (encrypted)	OS + JVM + application logs
Public IP	None	Private subnet -- ALB is the entry point
Process Mgmt	systemd service	Auto-restart on crash, start on boot

3. User Data Boot Script -- Steps

Step	Action
1	yum update -y -- patch all OS packages
2	Install Amazon Corretto 21 (Java 21)
3	Install CloudWatch Agent
4	Download ShopVerse JAR from S3 artifacts bucket
5	Fetch DB_HOST, DB_PORT, DB_NAME from SSM Parameter Store
6	Fetch DB_PASS, JWT_SECRET, REDIS_AUTH_TOKEN from Secrets Manager
7	Write /opt/shopverse/app.env (permissions 600)
8	Register shopverse systemd service
9	systemctl enable + start shopverse

4. IAM Instance Profile

Permission	Service Action	Purpose
S3 Read	s3:GetObject on artifacts bucket	Download ShopVerse JAR

Permission	Service Action	Purpose
SSM Read	ssm:GetParameter on /shopflow/*	Read config at startup
Secrets Manager Read	secretsmanager:GetSecretValue	Read passwords + keys
CloudWatch Logs	logs:PutLogEvents, CreateLogStream	Ship application logs
SSM Session Manager	ssmmessages:*, ec2messages:*	Secure terminal without SSH

5. SSM Session Manager vs SSH

Attribute	SSH (traditional)	SSM Session Manager (ShopFlow)
Port 22	Must be open in security group	Not needed -- no inbound ports
Key pairs	Must create and manage .pem files	Not needed -- IAM controls access
Private subnet	Requires bastion host	Works directly -- agent calls outbound
Audit trail	No native logging	Every session logged in CloudTrail

6. Network Placement

Component	Value
Subnet	shopflow-prv-1a (ap-south-1a private subnet)
Security Group	shopflow-ec2-sg -- port 8080 from ALB-SG only
Public IP	None
VPC	shopflow-vpc (10.0.0.0/16)